

Literature Review

Deep Learning-Based Battery Health Monitoring & RUL Prediction for AUVs

Problem Statement

scientific reports

OPEN

Remaining useful life prediction of lithium-ion batteries via spatial attention TLSTM and dilated CNN with evolutionary optimization

Suhua Chen^{1,2} & Xu Fang²

The precise estimation of the Remaining Useful Life (RUL) of lithium-ion batteries is essential for averting unforeseen failures and enhancing operational efficiency and maintenance planning. This paper presents an advanced deep learning framework that couples a spatial-attention mechanism with a Transductive Long Short-Term Memory (TLSTM) model, augmented by one-dimensional dilated convolutional layers to capture long-range temporal dependencies. In contrast to traditional LSTM or GRU models, our methodology utilizes one-dimensional dilated convolutional layers to effectively capture long-range temporal relationships and implements a clustering-based Differential Evolution

Chen & Fang, Scientific Reports

- Combines a spatial-attention mechanism with a Transductive LSTM (TLSTM) for RUL prediction
- Adds 1D dilated convolutional layers to capture long-range temporal dependencies, unlike plain LSTM/GRU
- Uses a clustering-based evolutionary (Differential Evolution) optimization step for tuning
- **Relevance:** shows how attention + dilated-conv layers can strengthen a recurrent RUL model – a useful extension beyond baseline LSTM/GRU



Article

A Lithium-Ion Battery Remaining Useful Life Prediction Model Based on CEEMDAN Data Preprocessing and HSSA-LSTM-TCN

Shaoming Qiu¹, Bo Zhang^{1,*}, Yana Lv¹, Jie Zhang² and Chao Zhang³

¹ Key Laboratory of Network and Communications, Dalian University, Dalian 116622, China; qiushaoming@dlu.edu.cn (S.Q.); lvyana@dlu.edu.cn (Y.L.)

² Beijing Jingwei Hirain Technology Company, Beijing 100020, China; zh929729285@163.com

³ ChengMai Technology (Nanjing) Company, Beijing 100080, China; zc2866128@163.com

* Correspondence: zhangbo@s.dlu.edu.cn

Abstract: Accurate prediction of the Remaining Useful Life (RUL) of lithium-ion batteries is crucial for reducing battery usage risks and ensuring the safe operation of systems. Addressing the impact of noise and capacity regeneration-induced nonlinear features on RUL prediction accuracy, this paper proposes a predictive model based on Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEMDAN) data preprocessing and IHSSA-LSTM-TCN. Firstly, CEEMDAN is used to decompose lithium-ion battery capacity data into high-frequency and low-frequency components. Subsequently, for the high-frequency component, a Temporal Convolutional Network (TCN) prediction model is employed. For the low-frequency component, an Improved Sparrow

Qiu et al., World Electric Vehicle Journal

- Decomposes battery capacity signal using CEEMDAN into high- and low-frequency components
- High-frequency component modeled with a **TCN**; low-frequency modeled with an Improved Sparrow-Search-optimized **LSTM**
- Directly targets noise and capacity-regeneration effects that hurt plain LSTM/TCN accuracy
- **Relevance:** closest methodological match to this project – same LSTM vs. TCN comparison, same capacity-regeneration challenge expected in CALCE data

Paper 3 – AUV Key Technologies Review & Summary



Ma et al., Engineering (Elsevier)

- Reviews AUV design, power systems, navigation, perception and AI applications
- Covers power-system challenges: energy density limits, deep-sea effects on battery performance and capacity estimation accuracy
- **Relevance:** domain motivation – explains why battery prognostics matters specifically for AUV endurance and reliability

Candidate methods shortlisted:

- Simple regression baseline
- LSTM, GRU (core recurrent models)
- TCN / CNN-LSTM hybrid
- Transformer (stretch work only)